[1] Anonymous Affiliation, City, Country (e-mail: anonymous@example.com)
Corresponding author: Anonymous (e-mail: anonymous@example.com).

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bullet.png Scientific question answering demands heterogeneous reasoning modalities: formula-driven numerical computation for physics and symbolic logical inference for formal logic. Deploying capable models under strict resource constraints—where only a 4B-parameter open-source base is available—requires co-design of the data curriculum, fine-tuning strategy, and reward specification. We present TRA-SAE (Training with Reward-Augmented Self-consistent Adaptive Evaluation), a three-phase pipeline for the EXACT 2026 competition that combines curriculum supervised fine-tuning (SFT) and Group Relative Policy Optimization (GRPO) with a novel unit-aware reward. Starting from Qwen3.5-4B against a public competition baseline of 38.71%, our best configuration achieves 53.92% overall accuracy (+15.21 pp), with 67.38% on physics and 28.95% on logic subtasks, using under $55 of A100 compute. We further show that self-consistency voting and Dual-LoRA expert routing do not reliably outperform single-adapter GRPO at 4B scale—a practical negative result for resource-constrained practitioners. Reward ablations reveal that a correctness-only signal converges faster at reduced training budgets, while structured auxiliary rewards yield greater benefit given sufficient training steps.

bullet.png Scientific reasoning, group relative policy optimization, curriculum fine-tuning, LoRA, physics question answering, logical reasoning, large language models, unit-aware reward

Large language models (LLMs) have demonstrated striking progress on mathematical and symbolic reasoning benchmarks [1]–[3]. Yet deploying competitive models in resource-constrained settings—where large proprietary APIs are unavailable or impractical—remains a significant practical challenge. Scientific question answering compounds this difficulty by requiring two qualitatively different reasoning modes within a single model: physics problems demand formula selection, dimensional analysis, and unit-consistent numerical computation, while logic problems require quantifier handling, propositional inference, and recognition of informal fallacies.

The EXACT 2026 competition provides a controlled testbed for this challenge, offering 1,945 labelled training examples across both domains and a hidden evaluation set with a published baseline of 38.71%. We treat this setting as a proxy for real-world mixed-domain scientific Q&A and investigate how far a Qwen3.5-4B (3.5B-parameter) open model can be pushed via publicly available fine-tuning techniques on a single A100 GPU.

Our pipeline, TRA-SAE, proceeds in three stages. First, a general curriculum SFT aligns the base model to a structured XML response format across all 1,945 training samples. Second, a logic-specialised SFT phase reinforces formal reasoning on the logic subset. Third, unit-aware GRPO applies group relative policy optimisation [4] with a composite reward that awards partial credit for physical unit consistency—a signal tailored to the dimensional nature of physics answers.

Our best configuration (cfg3) achieves 53.92% overall accuracy, outperforming the public baseline by +15.21 pp and all tested open-source baselines of comparable or larger size. A systematic ablation across six model configurations, six zero-shot baselines, and four reward variants characterises each component's contribution.

The main contributions of this paper are:

- A three-phase curriculum pipeline (general SFT $\rightarrow$ logic SFT $\rightarrow$ unit-aware GRPO) that achieves strong mixed-domain accuracy from a 4B base.
- A unit-aware reward function that awards partial credit for dimensionally consistent physical units, complementing correctness-based rewards.
- Negative results showing that self-consistency voting and Dual-LoRA expert routing do not improve over single-adapter GRPO at 4B scale.
- Reward ablations demonstrating that correctness-

only signals converge faster at reduced training budgets, while auxiliary rewards provide greater benefit with extended training.

Wei et al. [1] showed that step-by-step reasoning prompts substantially improve LLM performance on arithmetic and symbolic tasks. Wang et al. [3] extended this with self-consistency decoding: sampling diverse reasoning chains and marginalising over the most frequent answer. Our experiments show that self-consistency provides only marginal gains at 4B scale because the model's errors are often systematic rather than random, so majority voting reinforces rather than cancels errors.

DeepSeekMath [4] demonstrated that GRPO training on mathematical corpora reaches competitive performance without process-level supervision. Qwen2.5-Math [5] extended this to a 72B family with iterative self-improvement. Llemma [6] focuses on scientific and proof-oriented reasoning via continued pre-training on ArXiv and formal corpora. Our work differs in operating at 4B scale with a mixed-domain setting (physics and logic simultaneously) and introducing dimensionality-aware reward signals.

LoRA [7] reduces trainable parameters by decomposing weight updates into low-rank matrices, enabling full fine-tuning performance at a fraction of the cost. QLoRA [8] extends this with 4-bit quantisation. We apply LoRA ($r = 32$, $\alpha = 64$) to all seven projection layers, yielding ≈24M trainable parameters out of 3.5B. LoRAHub [9] motivates our Dual-LoRA routing configuration.

GRPO [4] simplifies PPO [10] by computing advantage estimates within a generation group, eliminating the need for a separate critic network. Neuro-symbolic approaches such as Logic-LM [11] and LINC [12] augment LLMs with formal solvers for verifiable logical reasoning; we integrate the Z3 SMT solver into our correctness reward.

Let $\mathcal{D} = \mathcal{D}_{\mathrm{phys}} \cup \mathcal{D}_{\mathrm{logic}}$ denote the EXACT 2026 dataset. Each sample $(q_i, a_i^*, s_i) \in \mathcal{D}$ consists of a question q_i, a ground-truth answer a_i^*, and a domain label $s_i \in \{\mathrm{physics}, \mathrm{logic}\}$. A model M_θ maps q_i to a free-text response $\hat{y}_i$ from which the predicted answer $\hat{a}_i$ is extracted. Evaluation accuracy on the validation split $\mathcal{D}_{\mathrm{val}}$ is:

$$\mathrm{Acc}(\theta) = \frac{1}{|\mathcal{D}_{\mathrm{val}}|} \sum_{i=1}^{|\mathcal{D}_{\mathrm{val}}|} \mathbf{1}[\mathrm{verify}(\hat{a}_i, a_i^*, s_i)], \quad (1)$$

where verify(·) performs Z3-based symbolic checking for logic problems and SI-unit normalised numerical comparison (≤ 5% relative tolerance) for physics problems. A structured XML response format is enforced to ensure reliable answer extraction, as described in Section IV-A.

All models—base and fine-tuned—respond with a three-tag XML structure: ⟨`reasoning`⟩ ⟨`answer`⟩ ⟨`explanation`⟩. This format facilitates reliable answer extraction, enables format-based reward, and encourages explicit chain-of-thought reasoning [1].

We fine-tune Qwen3.5-4B on all 1,945 training samples (1,213 physics, 732 logic) using cross-entropy loss on the target XML response. Hyper-parameters: 3 epochs, effective batch size 16 (batch 4, gradient accumulation 4), learning rate 2×10^{-4}, LoRA $r = 32$, $\alpha = 64$, dropout 0.05 applied to all seven projection matrices (Q, K, V, O, up, down, gate). Training converges in 58.8 min to loss 0.309 on one A100-80 GB GPU.

After Phase 1, we apply a second SFT pass restricted to the 732 logic training samples, with learning rate reduced to 10^{-4} for 1 epoch. This curriculum strategy allocates additional capacity to formal reasoning, which is underrepresented (37.6% of training data) yet requires distinct inference strategies compared with formula-based physics. Training converges in 15.7 min to loss 0.133.

Starting from the Phase 1.5 checkpoint, we apply GRPO [4] with a composite reward designed for mixed-domain scientific answers:

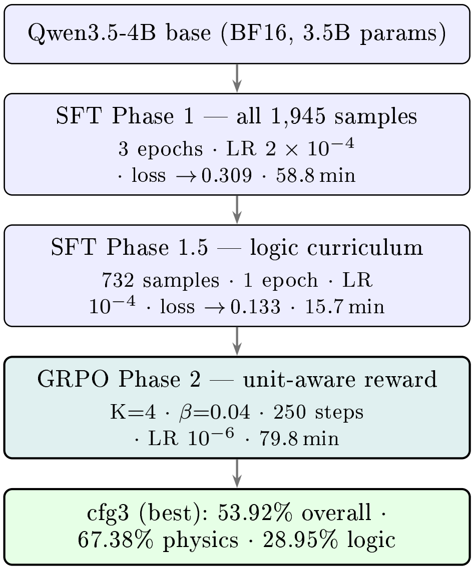
$$R(y, a^*, s) = w_f\, r_{\mathrm{fmt}}(y) + w_c\, r_{\mathrm{corr}}(y, a^*, s) + w_u\, r_{\mathrm{unit}}(y, a^*) + \lambda\, \mathbf{1}[l_y > \quad (2)$$

with w_f=0.30, w_c=0.60, w_u=0.10, λ=−0.10, where l_y is the reasoning token count.

Format reward r_{fmt}: awards 0.10 per XML tag present (⟨`reasoning`⟩, ⟨`answer`⟩, ⟨`explanation`⟩), maximum 1.0, scaled by w_f.

Correctness reward r_{corr}: binary (0 or 1) via domain-specific verification—SI-unit normalised numerical comparison for physics and Z3 solver [11] for logic.

Unit reward r_{unit} (novel contribution): for physics answers, r_{unit}=1 if the extracted answer contains a



Split	Total	Physics	Logic
Train	1,945	1,213	732
Val	217	141	76
Total	2,162	1,354	808

unit dimensionally compatible with the ground-truth unit under SI normalisation (e.g., `km/h` $\sim$ `m/s`), and 0 otherwise. This partial-credit signal penalises unit omission and dimensional errors independently of raw numerical correctness.

For each batch, GRPO generates K=4 completions per prompt and normalises rewards within each group before computing the policy gradient:

$$\hat{r}_k = \frac{R_k - \mu_R}{\sigma_R + \epsilon}, \quad k = 1, \ldots, K, \tag{3}$$

with KL penalty β=0.04, learning rate 10^{-6}, and 250 steps. Training converges in 79.8 min to loss 0.029.

We evaluate six configurations (Table 3): cfg0 is zero-shot; cfg1 applies SFT Phase 1 only; cfg2 adds SFT Phase 1.5; cfg3 (best) further adds GRPO Phase 2; cfg4 uses two separate LoRA adapters (physics and logic) routed by a TF-IDF + logistic regression classifier; cfg5 applies self-consistency voting (N=5) on top of cfg4.

Table 1 summarises the EXACT 2026 data split (90/10 stratified split by domain).

Parameter	Value
Base model	Qwen3.5-4B (BF16)
LoRA rank r	32
LoRA α	64
LoRA dropout	0.05
SFT learning rate	2×10^{-4}
SFT epochs (Ph. 1)	3
GRPO learning rate	1×10^{-6}
GRPO β (KL)	0.04
GRPO steps	250
GRPO generations K	4
Max new tokens	1,024
Reward w_f, w_c, w_u	0.30, 0.60, 0.10
Length penalty λ	−0.10 (if >800 tok)
GPU	A100-SXM4-80 GB

† <.

	System	Overall	Physics	Logic
Zero-shot baselines				
B1	Qwen3.5-4B	35.48	43.97	19.74
B2	Qwen2.5-Math-7B-Instruct	28.57	36.17	14.47
B3	Llemma-7B	12.90	5.67	26.32
B4	Mistral-7B-Instruct-v0.3	9.68	6.38	15.79
B5	Qwen2-Math-7B-Instruct	26.73	33.33	14.47
B6	DeepSeek-Math-7B-Instruct	11.98	9.93	15.79
TRA-SAE (fine-tuned Qwen3.5-4B)				
cfg0	Zero-shot	35.48	43.97	19.74
cfg1	+SFT Phase 1†	52.53	65.25	28.95
cfg2	+Logic SFT	51.61	65.25	26.32
cfg3	+GRPO (unit-aware)†	53.92	67.38	28.95
cfg4	Dual-LoRA + Router	49.77	60.28	30.26
cfg5	cfg4 + Self-Consistency	50.69	64.54	25.00

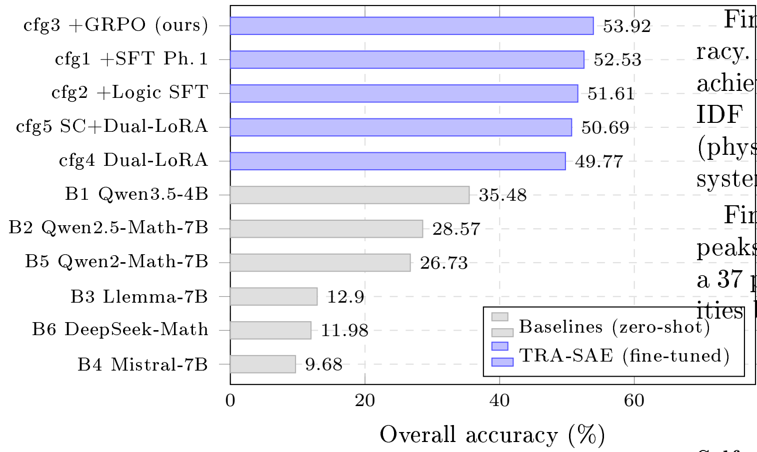
All figures are accuracy (%) on 217-sample val set.

We compare against six zero-shot baselines: B1 Qwen3.5-4B (our base, zero-shot); B2 Qwen2.5-Math-7B-Instruct [5]; B3 Llemma-7B [6]; B4 Mistral-7B-Instruct-v0.3; B5 Qwen2-Math-7B-Instruct; B6 DeepSeek-Math-7B-Instruct [4]. All baselines receive the same structured system prompt and are evaluated on the identical 217 validation samples under the verify metric of Eq. (1).

Table 2 lists the key hyperparameters.

Table 3 and Fig. 2 present the full comparison across baselines and TRA-SAE configurations.

cfg3 achieves 53.92% overall, surpassing the public competition baseline by +15.21 pp and outperforming every zero-shot baseline including larger 7B models. The dominant contribution comes from SFT Phase 1 alone (cfg0→cfg1: +17.05 pp), while GRPO adds a further +1.39 pp (cfg2→cfg3). Notably, Llemma-7B (B3) scores only 5.67% on physics despite strong logic



	Reward configuration	Overall	Physics	Logic
R1	Full: w_f=0.30, w_c=0.60, w_u=0.10, λ=−0.10	39.63	49.65	21.05
R2	No unit (w_u=0, w_c=0.70)	39.17	49.65	19.74
R3	No format (w_f=0, w_c=0.80)	38.25	46.81	22.37
R4	Correctness only (w_c=1.0)	41.47	51.06	23.68

All figures are accuracy (%).

performance (26.32%), confirming that mathematical pretraining does not automatically transfer to unit-constrained physics problems.

To isolate each reward term in Eq. (2), we retrain from the SFT Phase 1 checkpoint with four reward configurations, each for 150 GRPO steps (Table 4).

At the 150-step budget, the correctness-only variant (R4) yields the highest accuracy (41.47%). Removing the format reward (R3) hurts physics most (−2.84 pp vs. R1), confirming that structured output aids answer extraction. The unit reward's isolated contribution is modest at 150 steps (R1 vs. R2: −0.46 pp overall), yet it provides directional guidance that benefits longer training—the full reward at 250 steps in cfg3 reaches 53.92%.

Finding 1: Curriculum SFT provides the largest gain. The step from zero-shot (cfg0, 35.48%) to SFT Phase 1 (cfg1, 52.53%) adds +17.05 pp, dwarfing the GRPO contribution of +1.39 pp. At 4B scale, high-quality labelled supervision remains the dominant lever.

Finding 2: Format reward aids structured extraction. R3 (no format reward) drops physics accuracy by 2.84 pp relative to R1, indicating that explicit reward for XML compliance guides the model toward extractable answer formatting.

Finding 3: Dual-LoRA routing degrades overall accuracy. cfg4 and cfg5 both underperform cfg3, despite cfg4 achieving the highest logic accuracy (30.26%). The TF-IDF router misroutes ∼10% of border-zone questions (physics problems framed as conditionals), leading to systematic errors.

Finding 4: Logic remains a bottleneck. Logic accuracy peaks at 30.26% (cfg4) versus 67.38% for physics (cfg3), a 37 pp gap suggesting formal inference requires capabilities beyond standard SFT and GRPO at the 4B scale.

Self-consistency is effective when a model's errors are independently random across diverse reasoning chains [3]. At 4B parameter scale, errors are often systematic: the model consistently selects the wrong formula or mishandles the same quantifier across all N=5 samples. Majority voting then reinforces the error rather than cancelling it, yielding cfg5 (50.69%) below cfg3 (53.92%). Scaling to 7B+ models, where within-model diversity is greater, may recover the self-consistency benefit.

The reward ablation exposes a training-budget interaction: at 150 steps, the correctness-only reward (R4) converges fastest because the gradient signal is unambiguous—every parameter update is directly tied to answer correctness. The auxiliary rewards (r_{fmt}, r_{unit}) introduce weaker, more diffuse signals that require more steps to exert their cumulative benefit. The full reward formulation at 250 steps (cfg3: 53.92%) confirms this: given sufficient training, the multi-component reward yields a higher ceiling. Practitioners operating under tight compute budgets may therefore prefer a correctness-only reward, while those with more headroom benefit from the full formulation.

- Evaluation set size: 217 samples yield ±3 pp confidence intervals, limiting statistical power for distinguishing small gains.
- Single competition domain: Generalisation to broader benchmarks (SciBench [13], GPQA [14], OlympiadBench [15]) is untested.
- Router quality: TF-IDF routing at 90% accuracy introduces border-zone noise; a neural routing layer may help.
- Single training seed: Results are reported from single-run experiments; the large accuracy margins over baselines (>15 pp) suggest stable improvements, but multi-seed evaluation remains future work.

We presented TRA-SAE, a curriculum-driven pipeline for mixed-domain scientific reasoning that combines three-phase SFT with unit-aware GRPO. Starting from Qwen3.5-4B, our best configuration (cfg3) achieves 53.92% accuracy on the EXACT 2026 validation set—a +15.21 pp improvement over the public baseline—at a total compute cost of under $55 on a single A100-80 GB GPU.

The primary technical contribution is the unit-aware reward: by awarding partial credit for dimensionally consistent physical units, the reward function decouples unit errors from conceptual errors and provides sustained training signal beyond binary correctness. Our negative findings—that self-consistency voting and Dual-LoRA routing degrade accuracy at 4B scale—offer practical guidance for resource-constrained practitioners.

Future directions include: (i) scaling to 7B/14B checkpoints to recover self-consistency benefits; (ii) process-level reward via step-annotated data [16], [17]; and (iii) cross-dataset evaluation on SciBench and GPQA to assess generalisation beyond the competition setting.

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[1] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," Advances in Neural Information Processing Systems, vol. 35, pp. 24 824–24 837, 2022. [Online]. Available: https://arxiv.org/abs/2201.11903

[2] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," Advances in Neural Information Processing Systems, vol. 35, pp. 22 199–22 213, 2022. [Online]. Available: https://arxiv.org/abs/2205.11916

[3] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," arXiv preprint arXiv:2203.11171, 2023. [Online]. Available: https://arxiv.org/abs/2203.11171

[4] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo, "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," arXiv preprint arXiv:2402.03300, 2024. [Online]. Available: https://arxiv.org/abs/2402.03300

[5] A. Yang, B. Zhang, B. Hui, B. Gao, B. Yu, C. Li, D. Liu, J. Tu, J. Zhou, J. Lin, K. Lu, M. Xue, R. Lin, T. Liu, X. Ren, and Z. Zhang, "Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement," arXiv preprint arXiv:2409.12122, 2024. [Online]. Available: https://arxiv.org/abs/2409.12122

[6] Z. Azerbayev, H. Schoelkopf, K. Paster, M. D. Santos, S. McAleer, A. Q. Jiang, J. Deng, S. Biderman, and S. Welleck, "Llemma: An open language model for mathematics," arXiv preprint arXiv:2310.10631, 2024. [Online]. Available: https://arxiv.org/abs/2310.10631

[7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," arXiv preprint arXiv:2106.09685, 2022. [Online]. Available: https://arxiv.org/abs/2106.09685

[8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," Advances in Neural Information Processing Systems, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2305.14314

[9] C. Huang, Q. Liu, B. Y. Lin, T. Pang, C. Du, and M. Lin, "LoRAHub: Efficient cross-task generalization via dynamic LoRA composition," arXiv preprint arXiv:2307.13269, 2023. [Online]. Available: https://arxiv.org/abs/2307.13269

[10] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," in arXiv preprint arXiv:1707.06347, 2017. [Online]. Available: https://arxiv.org/abs/1707.06347

[11] L. Pan, A. Albalak, X. Wang, and W. Y. Wang, "Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning," arXiv preprint arXiv:2305.12295, 2023. [Online]. Available: https://arxiv.org/abs/2305.12295

[12] T. X. Olausson, A. Gu, B. Lipkin, C. E. Zhang, A. Solar-Lezama, J. B. Tenenbaum, and R. P. Levy, "LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers," arXiv preprint arXiv:2310.15164, 2023. [Online]. Available: https://arxiv.org/abs/2310.15164

[13] X. Wang, Z. Hu, P. Lu, Y. Zhu, J. Zhang, S. Subramaniam, A. R. Loomba, S. Zhang, Y. Sun, and W. Wang, "SciBench: Evaluating college-level scientific problem-solving abilities of large language models," arXiv preprint arXiv:2307.10635, 2024. [Online]. Available: https://arxiv.org/abs/2307.10635

[14] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman, "GPQA: A graduate-level google-proof q&a benchmark," arXiv preprint arXiv:2311.12022, 2023. [Online]. Available: https://arxiv.org/abs/2311.12022

[15] C. He, R. Luo, Y. Bai, S. Hu, Z. L. Thai, J. Shen, J. Hu, X. Han, Y. Huang, Y. Zhang, J. Liu, L. Qi, Z. Liu, and M. Sun, "OlympiadBench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems," arXiv preprint arXiv:2402.14008, 2024. [Online]. Available: https://arxiv.org/abs/2402.14008

[16] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," arXiv preprint arXiv:2305.20050, 2023. [Online]. Available: https://arxiv.org/abs/2305.20050

[17] P. Wang, L. Li, Z. Shao, R. X. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, and Z. Sui, "Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations," arXiv preprint arXiv:2312.08935, 2024. [Online]. Available: https://arxiv.org/abs/2312.08935